

Data Center Humidity Monitoring and Control System

Professional Instrumentation Report

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Introduction

Modern data centers house mission-critical hardware that is extremely sensitive to environmental conditions. Humidity, in particular, affects the lifespan of server components and the stability of electrical signals. This project implements a high-reliability monitoring and control system using the Rust programming language, focusing on ASHRAE standards and advanced computational numerical methods.

ASHRAE Standard Logic

The system is designed around the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) guidelines. The following thresholds govern the automated control plant:

- **Optimal Range (40% – 50% RH):** The target operating zone. No intervention required.
- **Marginal Zones (20–40% and 50–60%):** The system enters an alert state but continues monitoring before heavy intervention.
- **Critical High (> 60%):** Risk of liquid condensation and chemical corrosion. The system activates the dehumidifier immediately.
- **Critical Low (< 20%):** High risk of Electrostatic Discharge (ESD). The system activates the humidifier immediately.

System Architecture (Rust OOP)

The implementation utilizes a modular Object-Oriented approach in Rust to ensure memory safety and clear separation of concerns:

HumiditySensor Struct

Responsible for raw data acquisition. It includes methods for validation (rejecting values outside 0–100%) and recursive low-pass filtering to remove high-frequency noise.

Controller Struct

The logic core. It compares processed values against ASHRAE thresholds and generates control signals for the humidifier and dehumidifier actuators.

MonitoringSystem Struct

The orchestrator. It manages the data buffer for the moving average calculation and executes trend analysis through linear regression and root-finding algorithms.

Computational Numerical Methods

To move beyond simple threshold checking, the system applies the following mathematical models:

Moving Average

Calculates the mean of the sliding window buffer to smooth out transient spikes:

$$\text{Avg} = \frac{1}{n} \sum_{i=1}^n H_i$$

Linear Regression (Trend Analysis)

Calculates the slope of the humidity over time to predict if levels are rising or falling:

$$b = \frac{n \sum(xy) - \sum x \sum y}{n \sum(x^2) - (\sum x)^2}$$

Bisection Method (Root Finding)

Used to find the exact adjustment needed to return to the 45.0% target setpoint. The function $f(x)$ is defined as the deviation from the ideal:

$$f(x) = \text{Current Humidity} - 45.0 = 0$$

Flowchart Summary

1. **START:** Initialize system state and ASHRAE constants.
2. **INPUT:** Read raw data from the HumiditySensor.
3. **PROCESSING:** Apply Calibration → Filter → Moving Average.
4. **VALIDATION:** Check for sensor malfunction (out-of-bounds data).
5. **DECISION:** Evaluate ASHRAE thresholds.
6. **ACTUATION:** Toggle actuators based on the control loop results.

7. **MONITORING:** Display statistics and log trend analysis.
8. **LOOP:** Return to step 2.

Conclusion

By integrating numerical methods with professional industry standards, this system provides a robust solution for server protection. The use of Rust ensures that the monitoring software is free from common memory-related bugs, providing the high uptime required for modern data center operations.